

Rakesh Roy

Python / Django Backend Engineer

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Professional Summary

Python/Django Backend Engineer with 5+ years of experience building enterprise HRMS and SaaS applications, REST APIs, real-time systems, asynchronous processing, and database-backed business workflow solutions. Skilled in Django REST Framework, Django ORM, MySQL, Redis, Celery, and Django Channels/WebSockets, with hands-on exposure to cloud-based application deployment (AWS EC2/S3, DigitalOcean, Docker). Additional experience integrating Rasa for a proof-of-concept conversational interface.

Technical Skills

- **Languages:** Python, JavaScript, SQL, HTML5, CSS3
- **Backend:** Django, Django REST Framework, Django ORM, Django Channels, REST APIs
- **Databases:** MySQL, Redis
- **Async / Real-Time:** Celery, Redis, WebSockets
- **Cloud & DevOps:** AWS EC2, AWS S3, DigitalOcean, Docker, Git, CI/CD, Nginx, Gunicorn
- **AI / Conversational:** Rasa – Hands-on POC / Learning Integration
- **Data & Analytics:** Pandas, NumPy, Matplotlib, Plotly, Chart.js, Streamlit, Apache Superset, Bokeh
- **Authentication & Integration:** JWT, OAuth, Payment Gateway Integration (Payphi)
- **Frontend:** HTML, CSS, Bootstrap, JavaScript, jQuery, HTMX

Professional Experience

Medicover Hospitals

Python/Django Developer

Jul 2022 – Present

Hyderabad, India

- Developed and maintained a modular, multi-tenant HRMS currently serving 48+ clients, including RTC, covering recruitment, onboarding, attendance, leave management, TDS-enabled payroll, asset tracking, and performance appraisals
- Built RESTful APIs with Django REST Framework to support core HR business workflows across multiple client organizations
- Built real-time analytics dashboards using Streamlit and Django with Plotly/Chart.js visualizations for payroll trends and employee statistics
- Implemented Celery- and Redis-based scheduled reporting (daily/weekly/monthly/quarterly), generating and delivering payroll and HR reports across client organizations
- Developed real-time chat and notification capabilities using Django Channels and WebSockets to support internal HRMS communication
- Worked with AWS EC2 and S3 for application deployment and storage, gaining hands-on exposure to cloud-based application environments; used Docker and CI/CD pipelines to streamline deployments
- Integrated the Payphi payment gateway for secure fee/transaction processing within the platform
- Optimized MySQL database queries and implemented Redis caching to reduce response time and server load

Medicover Hospitals

Software Associate

Jul 2021 – Jun 2022

Hyderabad, India

- Developed core HRMS modules (attendance tracking, leave management, payroll processing) using Python/Django
- Built RESTful APIs with Django REST Framework, including authentication, serialization, pagination, and Swagger documentation
- Deployed and maintained applications on DigitalOcean with Nginx, MySQL, and automated backups
- Created dynamic Django views and templates for multi-tenant UI customization and organization-specific branding
- Implemented Git workflows, conducted code reviews, and participated in Agile ceremonies (sprint planning, standups, retrospectives)

- Completed a 6-month structured training program in Python, Django, and Django REST Framework under the mentorship of Mr. N. Naveen Kumar
- Built a full-stack web application demonstrating MVT architecture, Django ORM, authentication, and API development
- Gained hands-on experience in RESTful API design, database modeling, Git version control, and software development best practices

Key Projects

Multi-Tenant HRMS Platform

2021 – Present

Python, Django, Django REST Framework, MySQL, Redis, Celery, Django Channels, WebSockets, AWS

- Core platform architecture: multi-tenant SaaS HRMS with modules for recruitment, onboarding, attendance, leave, TDS-enabled payroll, asset management, and performance appraisals, now used by 48+ client organizations including RTC
- Background job pipeline built on Celery and Redis for scheduled report generation and delivery, decoupled from the main request/response cycle
- Real-time layer using Django Channels and WebSockets, enabling in-app chat and live notifications alongside the standard Django views
- Deployed on AWS (EC2, S3) with a Streamlit-based analytics frontend for KPI and payroll trend visualization

QR-Based Patient Token & Counter Management System

2022 – 2023

Python, Django, Django REST Framework, MySQL, Redis, Celery, Django Channels, WebSockets, AWS

- Built a patient token and queue management system generating QR-based tokens for patient identification
- Implemented token-to-counter mapping to route patients to the appropriate service counters/departments
- Designed the workflow to organize patient queue flow across multiple service counters

Rasa HR Chatbot – POC

2022 – 2023

Rasa, Python, Django

- Built a proof-of-concept conversational interface by integrating Rasa with the existing Django HRMS for learning and experimentation
- Configured intents, entities, and dialogue flows for sample HR use cases such as attendance, leave, and payroll queries
- Explored integration between the Rasa framework and Django HRMS APIs to handle sample conversational queries

Education

Bankatlal Badruka College For Information Technology (BBCIT)

Aug 2018 – Nov 2020

Master of Science in Computer Science

Hyderabad, India

- Relevant Coursework: Data Structures, Algorithms, Database Systems, Software Engineering, Web Technologies, AI/ML, Cloud Computing

Certifications & Awards

- **Certifications:** Python (Naresh I Technologies), Django (Satya Technologies), NCC 'B' Certificate – Grade A
- **Awards:** Gold Medal – Firing Competition (NCC ATC), 2nd Prize – Drawing Competition (NIC)